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Review

Review on Synthetic Approaches toward Rivaroxaban (Xarelto), an Anticoagulant Drug

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ABSTRACT: Venous thromboembolism including deep vein thrombosis and pulmonary embolism is an abundant and dangerous disease with significant relapse probability and mortality. In pursuit of an effective treatment, Bayer created Rivaroxaban, nowadays known under the brand name Xarelto, that was approved by the FDA in 2011. By 2022, the number of main indications for the drug has been increased 10-fold as well as the drug's prevalence in the field of thrombosis treatment. It is therefore apparent that with growing prescriptions new synthetic pathways are highly desirable, especially those utilizing inexpensive and safe reagents in environmentally benign protocols. The present work reviews the literature concerning Rivaroxaban preparation with an emphasis on the applicability of synthetic pathways on a large scale. Within the synthetic sequences, common intermediates are discussed in detail and relevant comparisons are made. Our main purpose is to provide an overview of the subject in its current state, paying attention to advances made and suggesting possible directions for further development.

KEYWORDS: *Rivaroxaban, Xarelto, factor Xa, anticoagulant, 4-(4-aminophenyl)morpholin-3-one, 5-chlorothiophene-2-carboxylic acid*

1. INTRODUCTION

Venous thromboembolism (VTE), which includes deep vein thrombosis (DVT) and pulmonary embolism (PE), has a frequency of 0.75−2.69 cases per 1000 people per year (and 2−7 for people older than 70), which makes it the third most abundant cardiovascular disease with myocardial infarction and stroke being the first and the second, respectively.[1,2] The economic burden to the healthcare system is also very significant, for example, for the treatment of VTE, the United States spends $7−10 billion a year,[3] while the European Union's expenditures are €1.5−2.2 billion at least.[4]

The two main types of VTE therapy are heparin (parenteral) and warfarin (oral), and both have issues with the dosage and patient response.[5,6] In addition, warfarin is known to interact with some other drugs[7] and even foods,[8] while heparin can cause an immune response, aggravating coagulation-related problems.[9]

To date, other oral anticoagulant drugs, such as Dabigatran, Rivaroxaban (**RIV**), Apixaban, and Edoxaban, have become available. These drugs have demonstrated superiority or at least noninferiority to heparin and warfarin while having a lower or comparable bleeding risk.[10] Their main advantages are the ease of dosage and the absence of the necessity of frequent laboratory monitoring.[11] The most widely used drug for the treatment of deep vein thrombosis and pulmonary embolism[12−14] is Rivaroxaban (brand name Xarelto). This drug, originally developed by Bayer, was approved by the FDA for use in knee and hip orthopedic surgery in 2011 (Figure 1), and since then the number of indications has increased 10-fold.[15]

Bayer (in Europe) and Johnson & Johnson (in the United States) are the two major manufacturers and suppliers of

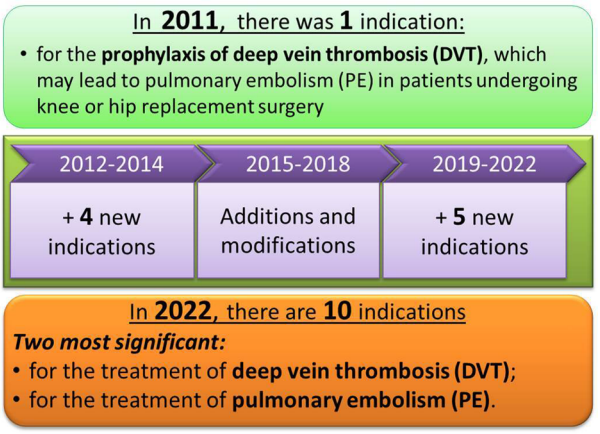
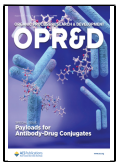


Figure 1. Timeline of the FDA approval of clinical indications.

Rivaroxaban. According to their financial statements,[16,17] the sales increased from €322 and $239 million in 2012 to €4.7 and $2.4 billion in 2021, highlighting growing usage of the medicine and giving an estimation for the current drug market. In addition, the latest three MEPS's surveys revealed the increase in the amount of Xarelto prescriptions from 6.5

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million in 2017 to 8.3 million in 2019 in the United States only.[18]

Mechanistically, Rivaroxaban is a reversible inhibitor of factor Xa, binding directly to its active site via S1 and S4 pockets,[19] with a 10 000 times higher selectivity than that for other biologically relevant serine proteases.[20] A comparison of action mechanisms for Rivaroxaban and several other anticoagulant drugs is given in Figure 2.[21] The central role

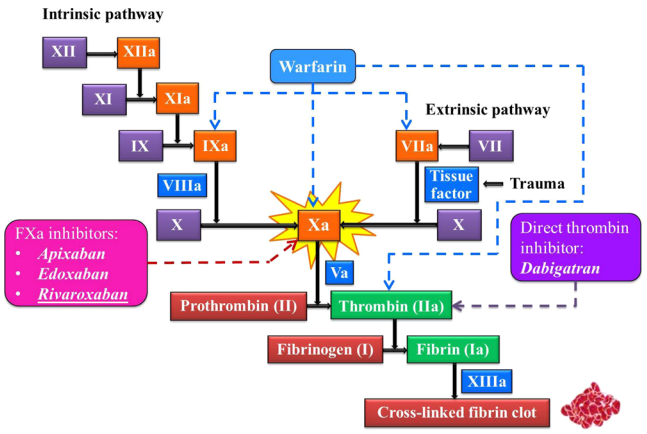


Figure 2. Blood-clotting cascade and anticoagulants' action mechanism.

in clot formation is played by thrombin (IIa), since it catalyzes the conversion of fibrinogen (I) into fibrin (Ia). Thrombin formation from prothrombin, in turn, is catalyzed by factor Xa. Therefore, to inhibit the fibrin clot formation, it is necessary to block either thrombin (IIa) or factor Xa. Warfarin is a nonselective inhibitor and interrupts the synthesis of factors IIa, VIIa, IXa, and Xa simultaneously, while more modern Dabigatran is selective to IIa inhibition, and Rivaroxaban, Apixaban, and Edoxaban are selective inhibitors of Xa.

It is interesting to compare the pharmacological characteristics for indirect (Warfarin) and direct (Dabigatran, Rivaroxaban, Apixaban, and Edoxaban) inhibitors.[22,23] The most striking difference between the two classes is the time required for full development of the anticoagulant effect (peak effect), which is noticeably lower for the latter (72−96 h for Warfarin vs 2−4 h for direct anticoagulants). In addition, within the direct anticoagulants group, Rivaroxaban possesses the highest bioavailability (>80%) and the lowest half-life period (5−9 h).

From the introductory section it should become clear that Rivaroxaban is among the most important anticoagulant drugs nowadays. Despite its presence on the market for about 10 years now, new synthetic routes for the drug are still under development in addition to the ongoing search for improvements for existing pathways. Therefore, the aim of the review is to discuss known synthetic routes and their merits and demerits and with all that data in hand find the best synthesis available so far and suggest possible directions for further developments. Thus, we decided to organize the review in the following way. First, a brief discussion of the Rivaroxaban structure and chemistry related to it is provided; then, a consideration of the key building blocks preparation is given, followed by an overview of the synthetic routes toward the target molecule. In many cases, we intentionally do not compare the whole pathways in terms of yields and scale since some steps are the same in ideology but done differently chemically. Instead, we try to analyze similarities from different synthetic sequences to highlight the best conditions for each step.

2. OVERVIEW OF SYNTHETIC ROUTES

From the structure of Rivaroxaban given in Figure 3 one notices the three main structural units: the *N*-arylmorpholin-3-

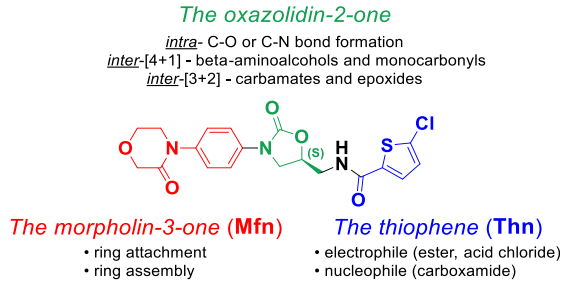


Figure 3. Structure of Rivaroxaban and its main constituents.

one (red), the 3,5-disubstituted 1,3-oxazolidin-2-one (green), and the 5-chlorothiophene-2-carboxylic acid derivative (blue). There are many ways to synthesize the drug under consideration. Since they utilize different chemistry, it is difficult to place all of the ideas in one scheme. However, some generalizations can still be made.

The morpholin-3-one building block preparations differed in the order of steps: in one strategy, the heterocyclic ring is attached to the aromatic one through either S_NAr or metal-catalyzed C−N bond formation; in the other, the morpholin-3-one moiety is assembled from an open chain precursor, attached to the aromatic system in advance.

The oxazolidin-2-one core synthesis was done in either an intra- or an intermolecular fashion. Suitable precursors for the first way are *N*-acylated *β*-aminoethanols and carbamates, possessing a leaving group at the *β* position to the oxygen atom; intermolecular [4 + 1] and [3 + 2] assembly was achieved using *β*-aminoethanols and one-carbon reagents (such as CDI), carbamates or isocyanates, and 1,2-difunctionalized compounds (such as epoxides, *β*-haloethanols, and *β*-sulfonatoethanols), respectively.

The sole thiophene-containing precursor is 5-chlorothiophene-2-carboxylic acid. In one strategy, this was used for the preparation of electrophilic derivatives (acid chloride, esters, or thioesters) as acylating agents for the nitrogen atom; in the other, the corresponding amide was prepared and employed as an *N*-nucleophile.

Rivaroxaban has one stereogenic center, and in all published cases, it came from a chiral starting material or a chiral synthesis rather than from resolution. Building blocks such as both isomers of epichlorohydrin, glycidol esters, or 3-functionalized propylene glycols are commercially available on any desired scale, while (*S*)-glycidyl phthalimide has to be prepared.

Despite the diversity of published syntheses, the majority of them use 4-(4-aminophenyl)morpholin-3-one and 5-chlorothiophene-2-carboxylic acid as starting materials. We believe that their preparations should be also included in this review as separate sections for the sake of completeness, while the oxazolidin-2-one ring syntheses are discussed where relevant. In addition, the synthesis of (*S*)-glycidyl phthalimide is also considered in its own section.

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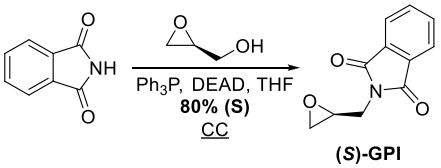
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3. BUILDING BLOCKS

To improve the information value of the reaction schemes, two things are used throughout the review. First, the "CC" acronym under the reaction arrows or near reaction yields means that column chromatography was used as a purification technique at a certain step, and second, the letters "S", "I", and "L" located near the yields indicate the average scale of the sequence: <200 mmol, S (for small); <2 mol, I (intermediate), and >2 mol, L (large). When scale data is not available in a publication, the n.s. designation is used. We believe that such enhancements would be valuable for the reader interested in a quick assessment of synthetic data.

3.1. (*S*)-Glycidyl Phthalimide, (*S*)-GPI. The desired compound was prepared (Scheme 1) via the Mitsunobu

Scheme 1. Synthesis of (*S*)-GPI via the Mitsunobu Reaction

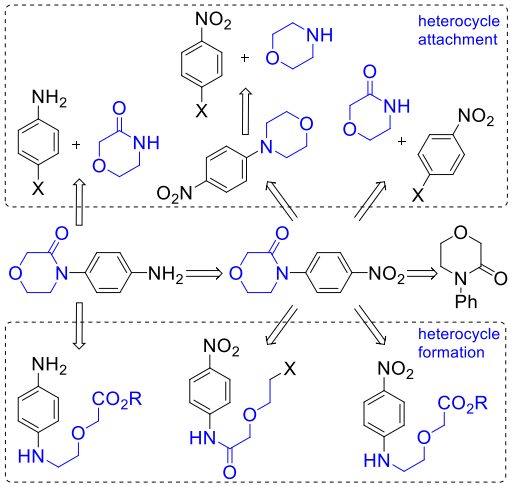


reaction from available (*S*)-(+)-glycidol and phthalimide.[24] Despite the good yield, stoichiometric byproducts triphenylphosphine oxide and the hydrazine from DEAD were generated, and chromatography was needed to achieve purification. Though the Mitsunobu reaction was employed successfully on a several kilogram scale (see, for example, a recent report[25]), it was done as a late-stage modification, probably justifying the wastes formed. Recently, a version of the Mitsunobu protocol, catalytic in both azo and phosphine reagents, was developed[26] as well as a fully organocatalytic one,[27] but to the best of our knowledge, none of these advances were used on a large scale. In view of this, it seems unreasonable to use the Mitsunobu route for such a small molecule as **(*S*)-GPI**, but the situation can change in the future.

Other references for approaches based on reactions of epichlorohydrin derivatives could be found elsewhere.[28] One interesting protocol (Scheme 2) employed on a very large (>100 kg) scale is noteworthy.[29] (*R*)-3-Chloropropane-1,2-diol reacted with potassium phthalimide to give diol intermediate **1** in excellent yield, and no purification was necessary. Next, three steps were conducted in one pot: the conversion of **1** into the corresponding dioxolane, then its ring opening into the acetyl chlorohydrin with TMSCl, and finally, the epoxide closure with sodium methoxide. The yield obtained might seem modest, but the ease of purification and relatively inexpensive precursors made the pathway worth consideration to say the least. The authors stated that no depression in the enantiomeric purity was observed during the sequence.

3.2. 4-(4-Aminophenyl)morpholin-3-one, MfnNH$_2$. A generalized scheme of the synthetic routes toward the key intermediate **MfnNH$_2$** is given in Scheme 3. The existing

Scheme 3. Overview of the Retrosyntheses of MfnNH$_2$

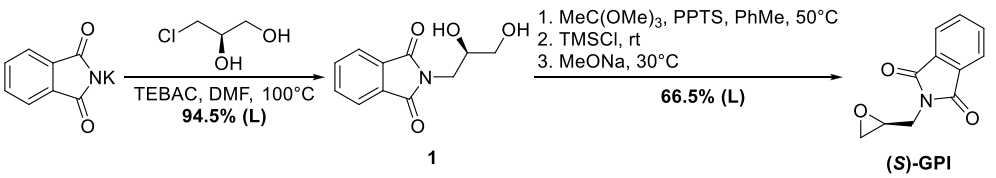


approaches can be subdivided into two major groups. The first one contains variants of the connection of morpholine (available) or morpholin-3-one (prepared separately) to the aromatic system. The second one includes examples of the assembly of the heterocyclic ring onto the aromatic one, preceded by the attachment of a suitable open-chain building block to it. It should be noted that these two approaches were applied to 4-(4-nitrophenyl)morpholin-3-one **MfnNO$_2$** as well.

3.2.1. Arylation of Morpholine/Morpholin-3-one. To achieve the assembly of the two rings, either morpholine or morpholin-3-one was reacted with a suitable aromatic electrophile (Table 1). Since morpholin-3-one itself is only weakly nucleophilic, its complete deprotonation with NaH was necessary to perform an S$_N$Ar reaction with 1-fluoro-4-nitrobenzene, but the product was obtained in only 18% yield (entry 1).[30] Other researchers attempted Ullmann–Goldberg (entry 2)[31] and Buchwald–Hartwig (entry 3)[32] aminations, but the results were far from promising. However, in the absence of the nitro group, the Ullmann–Goldberg reaction gave 4-phenylmorpholin-3-one **2** in a much higher 85% yield (entry 4).[31] There also were several attempts of copper-catalyzed morpholin-3-one arylation with 4-haloanilines (X = Br or I, Y = O, Z = NH$_2$),[33−35] but even on a small scale, the yields were about 50%, reaching 75% only under microwave irradiation.

Morpholine is much more nucleophilic than morpholin-3-one; thus, S$_N$Ar reactions with 1-halo-4-nitrobenzenes are relatively easy (X = F or Cl, Y = H$_2$, Z = NO$_2$). Such reports are numerous, and yields higher than 90% could be achieved when X = F or Cl.[36,37] Analogously, metal-catalyzed arylations

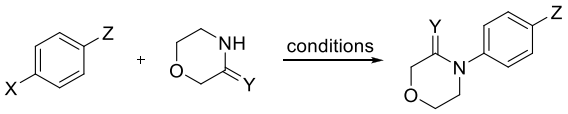
Scheme 2. Synthesis of (*S*)-GPI from PhthK and (*R*)-3-Chloropropane-1,2-diol



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Table 1. Synthesis of 4-Arylmorpholinones

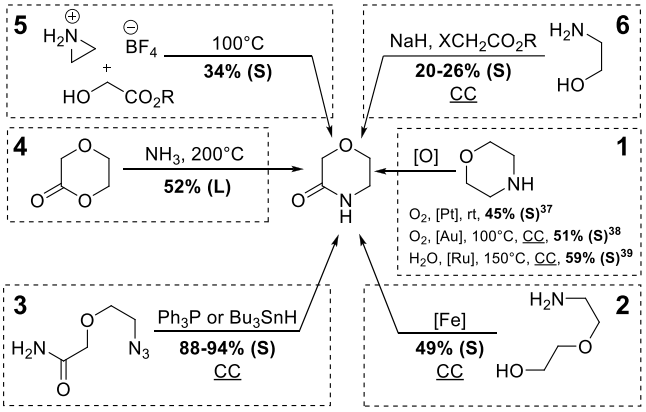


entry	X	Y	Z	conditions	product	yield (%)	ref
1	F	O	NO_2	NaH, NMP, rt	**MfnNO$_2$**	CC, 18 (L)	30
2	I	O	NO_2	Cu/C, Cs_2CO_3, BuCN, 120 °C	**MfnNO$_2$**	CC, 47 (S)	31
3	Br	O	NO_2	Pd(OAc)$_2$, XantPhos, Cs_2CO_3, dioxane, reflux	**MfnNO$_2$**	CC, n.s. (n.s.)	32
4	I	O	H	Cu/C, Cs_2CO_3, BuCN, 120 °C	**2**	CC, 85 (S)	31

of morpholine with 1-bromo- and 1-iodo-4-nitrobenzene are well known, but this approach is not discussed here since the use of the cheaper chloride without any metal participation gave satisfactory results.

3.2.2. Morpholin-3-one. Some syntheses discussed above used morpholin-3-one as a starting material. Despite the apparent simplicity of its structure, it is not readily available commercially and thus has to be synthesized. The two main pathways are oxidation of morpholine (Scheme 4, eq 1) and ring synthesis from suitable precursors (Scheme 4, eqs 2–6). As for the former, oxygen[38,39] (as an oxidant) and water[40] (as an oxygen source) were used to give the compound in moderate yields, but all of the reactions were tested only on a small scale and required precious metal catalysis. Moreover, column chromatography was necessary to achieve purification.

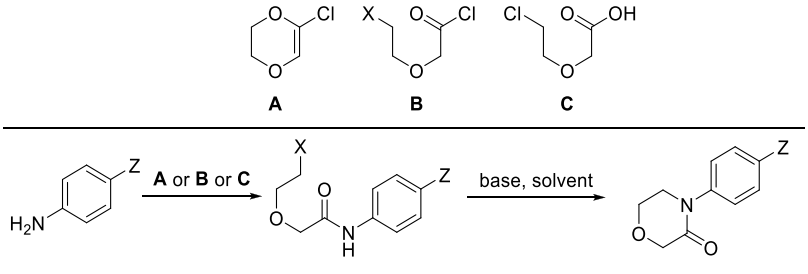
Scheme 4. Morpholin-3-one Preparation



The other group of approaches is ring formation. A dehydrogenative ring closure of an α,ω-aminoalcohol, catalyzed by a pincer-ligated divalent iron complex, was reported to yield 49% of the product (Scheme 4, eq 2).[41] Much higher yields were obtained[42,43] when an acyclic azide was converted into morpholin-3-one in Bu_3SnH- or Ph_3P-mediated processes (Scheme 4, eq 3), but in both cases, formation of stoichiometric byproducts makes the synthesis relatively unattractive, especially on a large scale. In the only scaled example of the target compound preparation[44] a high-temperature reaction between dioxanone and a large excess of ammonia was used (Scheme 4, eq 4). In addition to the moderate yield, the starting material is not very stable and the reaction requires an autoclave. The next two examples (Scheme 4, eqs 5 and 6) are low-yielding intermolecular alkylation–acylation sequences.[45−47] Thus, to the best of our knowledge, there is no reliable and efficient way to produce morpholin-3-one on a large scale from the available precursors. This fact implies that during the planning of a large-scale synthesis, the pathways including morpholin-3-one should be given a low priority.

3.2.3. 4-Arylmorpholin-3-one. Several related ways for morpholin-3-one ring construction have been reported. The general idea is to introduce an open-chain compound (either via S_NAr chemistry or acylation/alkylation) and then cyclize the products. 4-Nitroaniline and aniline itself (Scheme 5, Z = NO_2 or H) were acylated with (2-chloroethoxy)acetyl chloride (**B**, X = Cl)[48,49] or a very unusual reagent 2-chlorodioxene (**A**)[50] to afford chlorides **3** and **4** (Scheme 5) that cyclized upon treatment with a base. In the above-mentioned patent publication,[49] cyclization of related *O*-sulfonates **5** and **6** (Scheme 5) was reported to be effective as well, but the overall yields were somewhat lower than those for **3**. Acylation of 4-nitroaniline with (2-chloroethoxy)acetic acid (**C**) under boric acid catalysis was published,[51] but the yield of **4** was only 50%. A subsequent cyclization gave **MfnNO$_2$** in excellent yield, but

Scheme 5. 4-Arylmorpholin-3-ones Synthesis from Anilines



Z=H	**B** (X=Cl)	**3** (X=Cl, Z=H), **96%**	K_2CO_3, DMF	**2** (Z=H), **78%, Σ75% (I)**[49]
Z=NO_2	**A**	**4** (X=Cl, Z=NO_2)	K_2CO_3, MeCN	**MfnNO$_2$, Σ85% (S)**[50]
Z=H	**B** (X=OBs)	**5** (X=OBs, Z=H), **87%**	K_2CO_3, DMF	**2** (Z=H), **81%, Σ70% (I)**[49]
Z=H	**B** (X=OMs)	**6** (X=OMs, Z=H), **76%**	K_2CO_3, DMF	**2** (Z=H), **85%, Σ65% (I)**[49]
Z=NO_2	**C**	**4** (X=Cl, Z=NO_2), **50%**	NaOH, DCM-IPA	**MfnNO$_2$, 90%, Σ45% (I)**[51]

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Scheme 6. Morpholin-3-one Ring Synthesis from 1-Fluoro-4-nitrobenzene

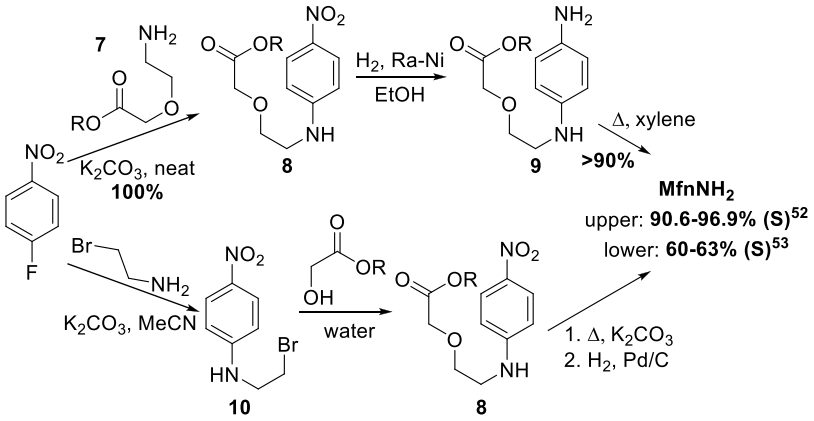
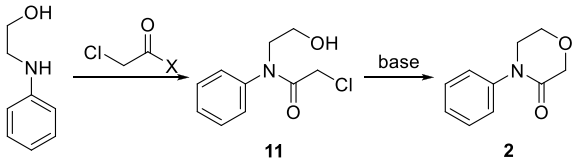


Table 2. Synthesis of 4-Phenylmorpholin-3-one (2) from 2-(Phenylamino)ethanol



entry	X	base	solvent	T (°C)	time (h)	yield (%)	ref
1	Cl	NaOH	EtOH−H_2O	n.s.	n.s.	80 (n.s.)	54
2	Cl	NaOH	EtOH−H_2O	38	1	89 (S)	55
3	Cl	NaOH	EtOH−H_2O	38−43	1	80 (L)	56
4	Cl	NaOH	EtOH−H_2O	38−45	1	68 (L)	57
5	OEt	*t*BuOK	THF	25	16	62 (L)	58

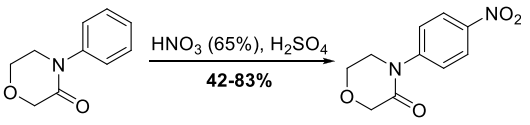
the sequence overall was significantly worse than one utilizing 2-chlorodioxene (**A**).

A different approach (Scheme 6) to the morpholin-3-one assembly was reported.[52,53] Both sequences started with an S_NAr reaction between a primary amine **7** and 1-fluoro-4-nitrobenzene. In the first case, the nitro group of **8** was hydrogenated and resulting compounds **9** cyclized on heating to give **MfnNH$_2$** in excellent yield. In the second variation, the product of the nucleophilic substitution **10** was used as an alkylating agent for an ester of hydroxyacetic acid to give the same substituted nitroaniline **8** as in the first synthesis. The cyclization under basic conditions followed by hydrogenation afforded **MfnNH$_2$** in moderate overall yield. In both publications, several hydroxyacetic esters were used, but as might be expected, the yields were virtually independent of it. In spite of the high yield provided by the former sequence, it should not be recommended for a large-scale application because of the first step. As shown by the authors, without an 8-fold excess of **7** the overall yield dropped below 60%.

Probably a more efficient (with respect to the availability of starting materials) approach (Table 2, entries 1−4) was used in the studies[54−57] where 2-(phenylamino)ethanol was acylated with chloroacetyl chloride (X = Cl) and intermediate **11** cyclized under basic conditions to give compound **2** in good to very good yields (68−89%). A similar approach[58] but with an ester of chloroacetic acid as an acylating agent (X = OEt) and a stronger base (*t*BuOK) gave the same product on a > 5 kg scale in 62% yield (Table 2, entry 5).

3.2.4. Nitration of 4-Phenylmorpholin-3-one (2). All of the literature examples[55−60] of the nitration of **2** (Table 3) followed essentially the same experimental procedure—slow addition of nitric acid to a solution of **2** in sulfuric acid usually

Table 3. Nitration of 4-Phenylmorpholin-3-one (2)



entry	HNO_3 (equiv)	H_2SO_4/HNO_3 (v/v)	T (°C)	time (h)	yield (%)	ref
1	1.6	5.8	−10	2	83 (S)	55
2	1.05	5.4	−5	2	70 (L)	56
3	0.98	5.0	−5	2	69 (L)	57
4	1.8	14.7	10−17	3	80 (L)	58
5	1.6	5.8	−10	2	83 (S)	59
6	1.6	6	−5	2	42 (S)	60

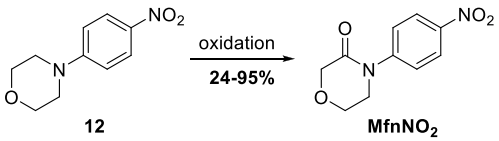
at 5−10 °C below zero. From the data of Table 3 it is clear that excessive amounts of nitric acid usually gave better yields (80−83%, entries 1, 4, and 5) than the stoichiometric loading (69−70%, entries 2 and 3). If a higher dilution was used (entry 4), the reaction could be conducted closer to rt with the same yield. If the lowest yield is discarded (entry 6), the average yield is 77%, and this value could be safely taken as an estimation of the scale-independent transformation's efficiency.

3.2.5. Oxidation of 4-(4-Nitrophenyl)morpholine (12). The first reported[61] small-scale oxidation procedure (probably an adaptation of the literature procedure[62] for a related substance) included the use of excessive $KMnO_4$, and the yield was less than 30% (Table 4, entry 1). Exactly the same protocol was attempted by Masse,[36] and he was able to raise the yield up to 39%, but once again the scale was small and purification was achieved only by column chromatography

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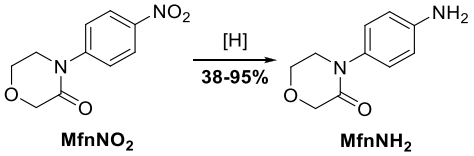
Table 4. Oxidation of 4-(4-Nitrophenyl)morpholine (12)



12 $MfnNO_2$

entry	oxidant	additive	solvent	T (°C)	time (h)	yield (%)	ref
1	$KMnO_4$	TEBAC	DCM	reflux	10	CC, 24 (S)	61
2	$KMnO_4$	TEBAC	DCM	reflux	15	CC, 39 (S)	36
3	$NaClO_2$	AcOH	MeCN−H_2O	50	2.5	91 (S)	37
4	$NaClO_2$	CO_2	MeCN−H_2O	50	5	95 (I)	64

Table 5. $MfnNO_2$ Reduction



$MfnNO_2$ $MfnNH_2$

entry	reductant	catalyst	solvent	T (°C)	time (h)	yield (%)	ref
1	H_2 (50 bar)	Pd/C	THF	70	8	38 (I)	30
2	H_2 (5 bar)	Pd/C	EtOH	80	1	93 (I)	56
3	H_2 (1 bar)	Pd/C	MeOH	rt	0.5	73 (S)	36
4	H_2 (1 bar)	Ra−Ni	MeOH	40−50	24	85 (I)	57
5	H_2 (3 bar)	Pd/C	THF	70	>6.5	89 (S)	59
6	H_2 (1 bar)	Pd/C	H_2O−AcOH	rt	n.s.	89 (S)	65
7	H_2 (2 bar)	Pd/C	MeOH	40	17	95 (L)	58
8	$N_2H_4{\cdot}H_2O$	Pd/C	EtOH	reflux	1	91 (S)	55
9	$N_2H_4{\cdot}H_2O$	$FeCl_3$	EtOH, MeOH	reflux	3, 5	91 (S, I)	37,49
10	$HCOONH_4$	Ra−Ni	H_2O	90	n.s.	87 (I)	66
11	$SnCl_2$		EtOAc−EtOH	reflux	2	92 (S)	61
12	Fe		AcOH	30	3	95 (I)	67
13	$Na_2S_2O_4$		MeOH−H_2O	50	5	93 (S)	68

(Table 4, entry 2). Recently, another oxidant was suggested[63] for the transformation: a rather complex mixture of $NaClO_2$, NaOCl, and NaH_2PO_4 under TEMPO catalysis gave related 4-phenylmorpholin-3-one (**2**) in 90% yield. That finding inspired Liu et al.[37] to further optimize the oxidation conditions, and it was found that the use of $NaClO_2$ alone gave **$MfnNO_2$** in 91% yield (Table 4, entry 3). To avoid solid wastes from NaH_2PO_4 used previously, the authors utilized acetic acid for a pH adjustment. In a very recent work,[64] the same group showed that the desired solution acidity could be achieved by conducting the oxidation under a CO_2 atmosphere. The product yield under the new conditions was as high as 95% on 200 mmol scale (Table 4, entry 4).

*3.2.6. Reduction of **$MfnNO_2$**.* Reduction of the nitro group of **$MfnNO_2$** is the last step in many synthetic routes toward **$MfnNH_2$**. One of the easiest ways to reduce an aromatic nitro compound is heterogeneous catalytic hydrogenation, so it is not surprising that many attempts were undertaken (Table 5, entries 1−7). Most hydrogenations were carried out at low hydrogen pressure and elevated temperatures, giving the product in good to excellent yields. Palladium on carbon was widely employed as a catalyst, while Raney nickel usage was relatively rare. A solvent seems to play only a minor role in the overall process. Entry 7 in Table 5 corresponds to the largest scale hydrogenation, and its results therefore could be considered as the most significant and reliable.

Apart from molecular hydrogen, several transfer hydrogenation systems (Table 5, entries 8−10) as well as ordinary inorganic reductants (Table 5, entries 11−13) were successfully utilized. Though the product yields obtained with hydrogenation and non-hydrogen reductants are essentially in the same range, the former way is preferred because it is significantly more environmentally benign. Inorganic wastes are inevitably generated when metals or salts are used in addition to chemoselectivity issues that were found to occur with hydrazine as a hydrogen source.[37]

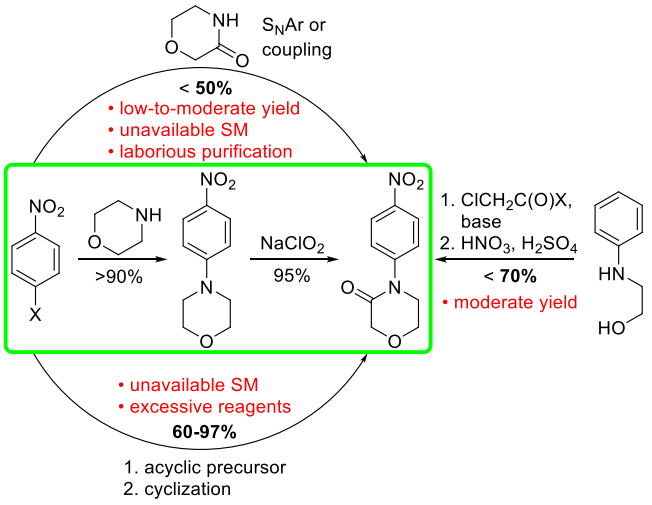
From this section of the review, it is clear that **$MfnNO_2$** can be synthesized in a number of ways (Scheme 7). However, the scalability for most of them was not tested, and the conditions were not optimized. Therefore, we want to discuss some points and highlight the most promising synthetic steps.

First, upon comparing reported data it can be concluded that the introduction of a morpholine moiety via S_NAr chemistry provides high yields, utilizes reasonable reaction conditions (refluxing temperature of morpholine for a few hours, the possibility to conduct the reaction without the solvent), and available starting materials (e.g., 1-chloro(fluoro)-4-nitrobenzene and morpholine). In addition, the purification can be achieved by simple dilution of the reaction mixture with water and filtration of the solid. No thorough time- and energy-consuming drying is necessary, since the next stage (oxidation) is conducted in a water-containing mixture of solvents. Despite the synthesis with separately prepared morpholin-3-one becomes "more" convergent, it suffers from the lack of any large-scale acceptable routes toward morpholin-3-one itself and difficulties with its connection to the aromatic

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Scheme 7. Summary of Pros and Cons of Different Synthetic Pathways to $MfnNO_2$



ring. Obvious problems, associated with the assembly of the heterocyclic ring onto the aromatic one, are the necessity to use unavailable open-chain precursors (sometimes in a large excess) and moderate yields of the product, i.e., in the case of 2-(phenylamino)ethanol.

Second, oxidation of **12** has to be conducted next. According to the literature data, a perfect procedure is that using sodium chlorite under CO_2 atmosphere. This oxidant and the MeCN solvent are inexpensive and relatively green, and the yield is excellent. Due to the simplicity of the experimental setup, a low reaction temperature, and its homogeneity, there is little doubt in the scalability of the conditions. However, further scaling of the oxidation reaction requires additional safety considerations.[69] We also believe that this stage is preferred to the nitration of **2**, obtained through the above-mentioned ways.

The last step of the **$MfnNH_2$** synthesis is reduction of the nitro group. The majority of literature sources used catalytic hydrogenation, and despite other reductants being reported, we think the catalytic hydrogenation is the method of choice, especially on a large scale. The conditions in entry 7 of Table 5 seem to be near optimal and hardly require any significant adjustment.

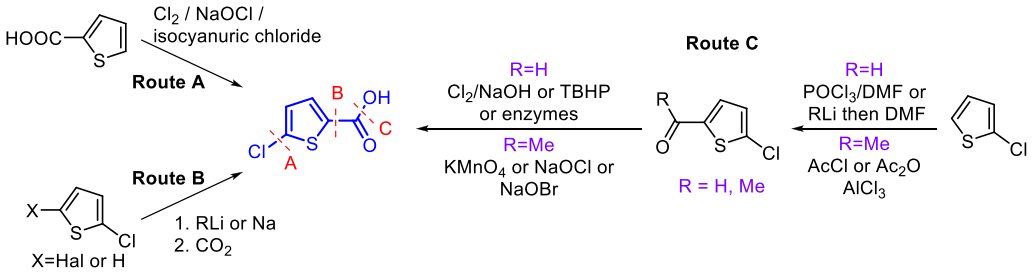
3.3. 5-Chlorothiophene-2-carboxylic Acid, ThnOH. The straightforward one-stage ways to the target compound are chlorination[70−72] of thiophene-2-carboxylic acid (Scheme 8, Route A) or carboxylation of 2-chlorothiophene (Scheme 8, Route B, X = H),[73−76] since both starting materials are readily available. Direct halogenation of thiophene-2-carboxylic acid with chlorine was reported[70] as early as 1934, but no yield was given. Another preparation of the title compound was conducted[71] with NaOCl as the chlorinating agent, and a 46% yield was obtained on a small scale. Recently, chlorination of the acid with isocyanuric chloride under three sets of conditions was published.[72] In the first case, the reaction was carried out in concentrated sulfuric acid to give **ThnOH** (82.2%, I), while in the other two acetic acid was used as a solvent (72.9%, I) or a cosolvent (70.0%, I).

The first report on a successful carboxylation of 2-chlorothiophene was made in 1948.[73] 5-Chloro-2-thienylsodium was generated in situ from 2-chlorothiophene and sodium shot (a form of highly dispersed sodium prepared by slow cooling of a vigorously stirred molten sodium emulsion in toluene or xylene) and quenched with dry ice to give **ThnOH** in 70% (I) yield. A similar batch procedure but with nBuLi as a base gave a comparable yield of 65% (S)[74] with dry ice. Continuous flow syntheses with gaseous CO_2 were also reported to give >70% (n.s.)[75] and 77% (S)[76] yields of **ThnOH** with nBuLi and LDA as bases, respectively. Ag-mediated C−H activation/carboxylation is known,[77] but at present, the procedure is more of laboratory interest. In addition to metalation of 2-chlorothiophene, metal−halogen exchange chemistry of 2,5-dichlorothiophene and 2-bromo-5-chlorothiophene (Scheme 8, Route B, X = Hal) was employed successfully,[74,78] but the approach is obviously less atom economic than the deprotonation−carboxylation sequence. In addition to the methods, enhancing the thiophene ring nucleophilicity, a different strategy, utilizing electrophilic activation of CO_2 with $EtAlCl_2$, was reported[79] to give the acid in only 41% (S) yield.

ThnOH was obtained through oxidation of the corresponding aldehyde or ketone (Scheme 8, Route C). 5-Chlorothiophene-2-carboxaldehyde was oxidized with Cl_2/NaOH[80] (<76%, L) and TBHP (96%, S)[81] and enzymatically (99%, S).[82] The yields were excellent, and scalability was proven in the first case, but the aldehyde itself is not readily available. Formylation of 2-chlorothiophene was done through the Vilsmeyer−Haack chemistry (formamides/$POCl_3$)[83−86] with yields up to 87% and via a metalation−acylation sequence.[87,88] These pathways are summarized in Scheme 8 (Route C, R = H).

Oxidation of ketones to form carboxylic acids requires carbon−carbon bond cleavage and is usually done with methyl ketones to waste no more than one carbon atom. The feasibility of **ThnOH** preparation through oxidation of 2-acetyl-5-chlorothiophene was shown in 1886[89] using alkaline $KMnO_4$ as an oxidant, but no yield was given. After that, sodium hypochlorite (92%)[90] and sodium hypobromite (61−76.1%, I, S)[91,92] were used as oxidizing agents (via the haloform reaction). **ThnOH** was also obtained from "prehalogenated" 5-chloro-2-(trichloroacetyl)thiophene by

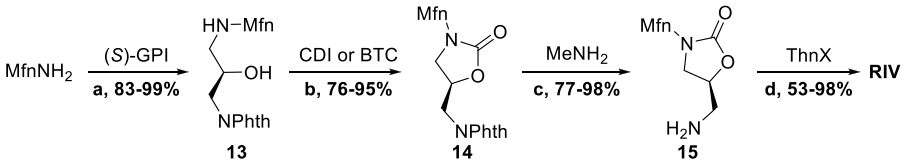
Scheme 8. Overview of Synthetic Pathways to ThnOH



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Table 6. Syntheses of RIV with (*S*)-GPI as a Chiral Building Block



entry	a	b	c	d	yield (%)
1	IPA−H_2O, reflux 82.6%	K_2CO_3, DCM, rt 88.2%	MeOH, reflux 80%	X = Cl, K_2CO_3, DCM, 88% X = 4-$NO_2C_6H_4O$, Et_3N, DMSO, 82.7%	48−51 (I)[102]
2	MeOH, 60 °C 82.7%	PhCl, 100 °C 93.8%	EtOH−H_2O, 60 °C 76.9%	X = Cl, Et_3N, DCM, 90.2%	54 (L)[103]
3	H_2O, 80 °C 98.7%	DMAP, DMC, 80−85 °C 89%	H_2O, 35 °C	X = Cl, K_2CO_3, DMC, 64%	56 (I)[66]
4	EtOH−H_2O, reflux 92%	DMAP, THF, 60 °C 87%	EtOH, reflux 98%	X = Cl, Py, 0 °C to rt, 86%	67 (S)[19]
5	EtOH−H_2O, 65 °C 97.2%	DMAP, THF, 60 °C 87%	EtOH, 70 °C 86%	X = Cl, Et_3N, EtOAc−H_2O, 85%	62 (S)[104]
6	IPA−H_2O, reflux 96%	K_2CO_3, DCM, 25−30 °C 95%	MeOH, 60−65 °C 85%	X = 4-$NO_2C_6H_4O$, Et_3N, DMSO, 86%	67 (I)[51]
7	MeOH−H_2O, 65−70 °C 92%	Et_3N, DCM, 0−5 °C 89%	IPA, 45−50 °C 94%	X = Cl, Na_2CO_3, Me_2CO, 98%	75 (I)[105]
8	H_2O, 55−60 °C 85%	DMAP, THF, 55−60 °C 76%	MeOH, 60−65 °C 94%	X = OH, CDI, HOBT, THF, 53% X = OTs, THF, 64%	32−39 (I)[106]

alkaline hydrolysis in 95% (S) yield.[93] The starting acetylthiophene itself is readily available via an ordinary Friedel−Crafts procedure in an excellent yield of 94%.[94] The data are summarized in Scheme 8 (Route C, R = Me).

In almost all cases, the thiophene building block is installed into Rivaroxaban through the corresponding acid chloride **ThnCl**; it was prepared from the acid using conventional $SOCl_2$[91,95] or $(COCl)_2$[96,97] or from 2-acetyl-5-chlorothiophene with a somewhat exotic reagent S_2Cl_2 (68%).[98] **ThnCl** was usually used without any purification, except for the removal of a chlorinating agent excess (if any).

In summary, several routes toward **ThnOH** and its acid chloride **ThnCl** have been described in the literature. Probably the most promising are those utilizing carboxylation of a metalated 2-chlorothiophene derivative with CO_2 since it is more atom economic, generates less wastewater, and provides higher yields than formylation/oxidation or acylation/oxidation sequences. The metalations were carried out in ether or THF, but utilizing greener 2-MeTHF seems very probable in light of recent developments in the field (see recent reviews[99−101] and references therein). As for the formation of the acid chloride, the use of thionyl chloride is preferred because it is cheaper than the alternatives and gives very high yields. Toluene, benzene, and chlorinated hydrocarbons are common solvents for the transformation, and on a large scale, the preference should be given to the former.

4. COMPARISON OF FULL SYNTHETIC SEQUENCES

The present section provides a comparison of whole synthetic sequences leading to Rivaroxaban. Where possible, several protocols are generalized on one scheme to avoid any unnecessary redundancy and simplify comparisons. The synthetic pathways are grouped according to the chiral building block used.

4.1. (*S*)-Glycidyl Phthalimide, (*S*)-GPI. The vast majority of the Rivaroxaban preparations that include (*S*)-**GPI** as a chiral building block followed the synthetic sequence given in Table 6.[19,51,66,102−106] The syntheses started with the ring opening of (*S*)-**GPI** (step a) followed by oxazolidin-2-one **14** formation via a reaction between substituted 2-aminoethanol **13** and a one-carbon electrophile, such as CDI (1,1′-carbonyldiimidazole) or BTC (bis(trichloromethyl) carbonate, triphosgene) (step b). The phthalimido protecting group was removed upon treatment with an excess of aqueous methylamine (step c), and amine **15** thus obtained was usually isolated as its salt. The last step d is acylation, which is commonly conducted using preformed **ThnCl**, but in a few cases, other approaches were employed. From Table 6, it is clear that most stages could be carried out in excellent yields, but the outcome depends upon the exact conditions used.

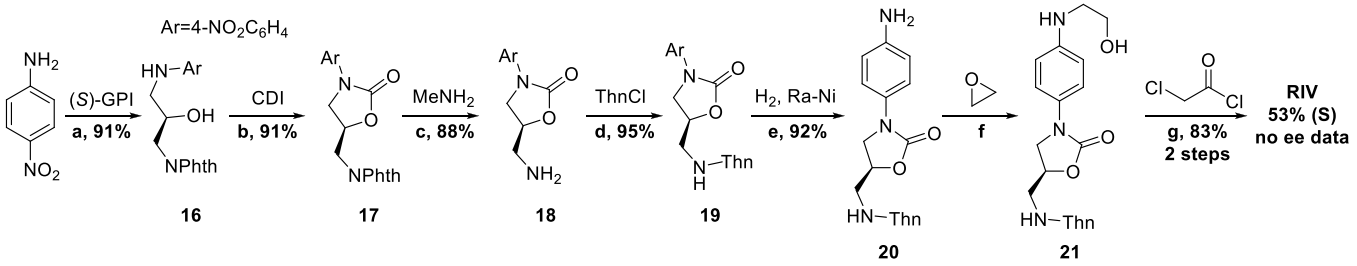
Step a was always conducted in alcoholic solvents with or without added water or in water itself at temperatures that were close to the boiling points of the reaction mixtures. There is no apparent relationship between the main reaction variables, such as solvent, temperature, and time, and the product yield. In many cases, an equimolar or slightly superstoichiometric amount of (*S*)-**GPI** was utilized, but sometimes the use of more than a 50% excess of the reagent was described (entries 3 and 5).[66,104] In all cases but one (entry 4),[19] product isolation was achieved by filtration of the reaction mixture cooled to rt or somewhat below.

Acylation of **13** with CDI or BTC (step b) usually gave high and similar yields (if the lowest value is omitted), which makes it very difficult to distinguish important reaction parameters. Up to a 4-fold excess of CDI was used, while the range of 1.5−2 equiv was the most common. One synthesis (entry 2)[103] was conducted with only 1.2 equiv of CDI at 100 °C to furnish the product in 93.8% yield in 4 h. Lower temperatures resulted in a significant rate decrease: 6 h at 80−85 °C (entry 3)[66] and 24 h at 60 °C (entry 4)[19] were required to complete the reaction even with larger amounts of CDI. To improve the procedure (entry 2), replacement of chlorobenzene with less harmful toluene could be suggested.

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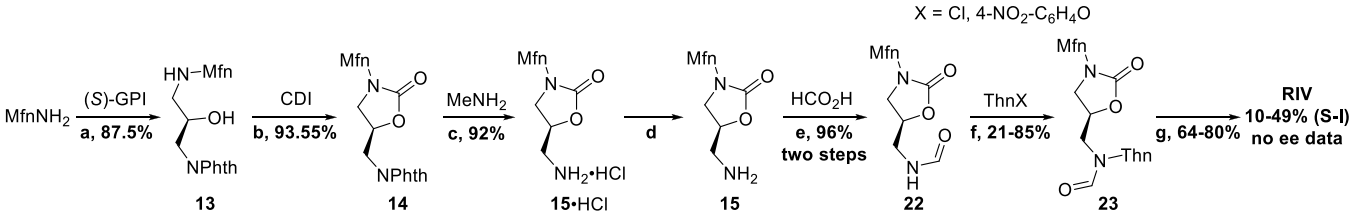
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Scheme 9. Synthesis of Rivaroxaban from 4-Nitroaniline and (*S*)-GPI[a]



[a]Conditions: (a) EtOH, 50 °C. (b) DCM, 20 °C. (c) IPA, 50 °C. (d) Et_3N, DCM, 0−10 °C then rt. (e) MeOH, 40 °C. (f) IPA, 60 °C. (g) NaI (cat.), EtOH−H_2O, 60 °C then 45% aq. NaOH, 40 °C.

Scheme 10. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-GPI[a]



[a]Conditions: (a) IPA−H_2O, reflux, 24 h. (b) K_2CO_3, DCM, rt, 5 h. (c) (i) MeOH, 60−65 °C, 4 h; (ii) HCl, pH = 1−2. (d) K_2CO_3, EtOH, rt, 2 h. (e) PhMe, 110−120 °C, 3−4 h, azeotropic water removal. (f) K_2CO_3 or NaOH or DIPEA, DCM, or DCM−H_2O, rt or reflux, 1−12 h. (g) AcOH−HCl, 70−80 °C, 3−4 h or 70% MsOH or HCl or H_2SO_4, DCM, reflux, 4−5 h.

The phthalimido group removal (step c) was achieved using an excess of aqueous $MeNH_2$ at elevated temperatures. In most cases, isolation of amine **15** was done through its salts, and the yields were close to 90% with one exception: when nitric acid was used, the yield dropped below 80%.[103] The highest yield of the salt was obtained with perchlorate as a counterion.[105]

The amide formation in step d was frequently conducted with **ThnCl**,[19,66,102−105] but esters[51,102,106] and **ThnOH** itself[106] were also used. As a rule of thumb, the reactions with **ThnCl** gave better (85−98%) results than the other derivatives (53−86%).

The overall yields obtained were between 32% and 75%, while the three best syntheses gave 65−75% (entries 4, 6, and 7).[19,51,105] One of them is an early scientific paper,[19] where the authors operated on a small scale and all isolations were conducted by means of column chromatography, which is unlike the other two works[51,105] with no chromatography and higher loadings. These two syntheses differ in steps b and d: the former used CDI and an ester of **ThnOH**, while the latter employed BTC and **ThnCl**. The BTC reagent, though readily available and cheap, requires special safety precautions associated with the phosgene toxicity[107] and thus should not be recommended for large-scale applications. The use of CDI is much easier and safer because it generates only innocuous products upon reaction or hydrolysis and does not require special handling. As for the acylation step, the use of **ThnCl** rather than an ester is significantly more atom economic in addition to the simpler preparation of the former.

Unfortunately, not all publications provided enantiomeric purities of the products, but those that did reported similar and high ee values of 99−99.99%. Since the chirality was introduced by the same reagent under similar conditions, it is reasonable to anticipate that for the sequence in Table 6, the unreported ee values were in the same range.

It is noteworthy that the synthesis given in Table 6 was once attempted in one pot to give Rivaroxaban in 52% yield,[66] while the "stepwise" yield obtained by the same group was 56% on the same scale. The difference of only 4% is rather insignificant and implies that one-pot versions of higher yielding protocols could also be developed in the future.

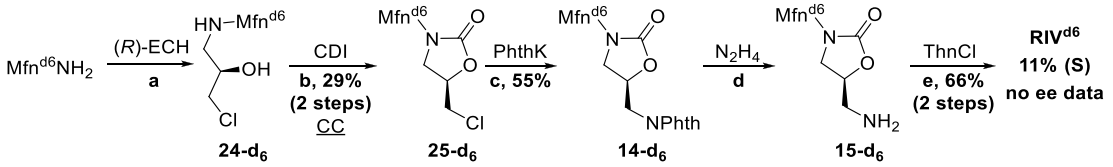
The sequence in Scheme 9[108] started with 4-nitroaniline rather than **$MfnNH_2$**, but the first four steps were chemically the same as those in Table 6: the ring opening gave *β*-aminoethanol **16** that was reacted with CDI (to oxazolidin-2-one **17**), deprotected (to primary amine **18**), and, finally, acylated with **ThnCl** to give intermediate **19**. This was hydrogenated over Raney nickel, and aniline **20** thus obtained was reacted with ethylene oxide to give the corresponding 2-aminoethanol **21**, which was further converted into Rivaroxaban upon treatment with chloroacetyl chloride and a base. To compare this pathway with the one in Table 6, one needs to estimate the yield of the **$MfnNH_2$** preparation from 4-nitroaniline. In our opinion, the optimal synthesis of **$MfnNH_2$** has three steps from 1-chloro-4-nitrobenzene, and the overall yield is greater than 80% (see section 3.2). Though not rigorous, both nitrobenzenes are cheap bulk chemicals, and the yield from different starting material could be compared. Taking this value into account, the sequence in Table 6 becomes a 7-step procedure with an overall yield of 26−60%. Although the synthesis in Scheme 9 seems to be similar to the best examples of Table 6 and has the same number of steps, the usage of hazardous and toxic oxirane and possibly pyrophoric Ra−Ni outweighs the high yield.

The first three steps depicted in Scheme 10,[109] are the same as those in Table 6. Then, amine **15** was converted into its formyl derivative **22** with subsequent acylation and deformylation. Several sets of conditions were tested for the acylation step, and it was found that **ThnCl** and DIPEA as a base gave the desired imide **23** in 85% yield. The deformylation step was optimized as well, but the highest yield obtained was only 80%. An attempt was made to synthesize the desired drug directly from **15** using 5-chlorothiophene-2-carbonitrile as an acylating

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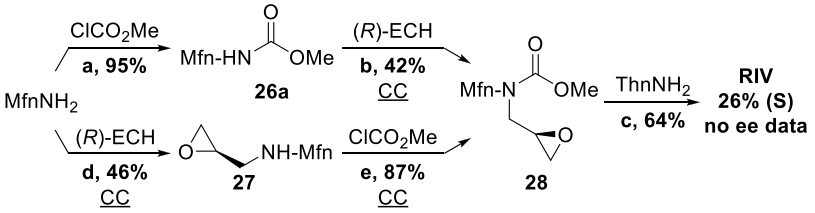
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Scheme 11. Synthesis of Deuterated Rivaroxaban from $MfnNH_2$ and (*R*)-ECH[a]



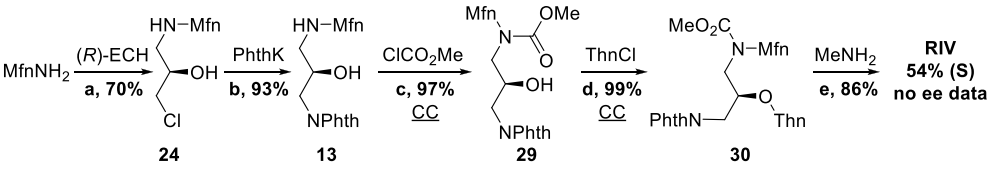
[a]Conditions: (a) IPA, reflux, 16 h, N_2. (b) DCM, rt, 3 h, N_2. (c) DMF, 120−125 °C, 8 h, N_2. (d) EtOH, reflux, 4.5 h. (e) Et_3N, DCM, 0 °C to rt, 16 h.

Scheme 12. Synthesis of Rivaroxaban from $MfnNH_2$ and (*R*)-ECH[a]



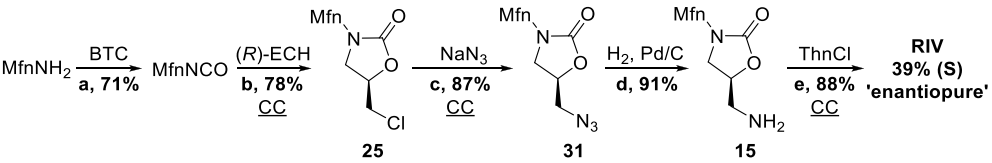
[a]Conditions: (a) DIPEA, DCM, rt, 1 h. (b) NaH, DMF, 60 °C, 3 h, N_2. (c) LiHMDS, THF, reflux, 3 h, N_2. (d) (i) IPA, reflux, 20 h (ii) $NaHCO_3$, reflux, 1.5 h. (e) DIPEA, MeCN, 0 °C to rt, 30 min.

Scheme 13. Synthesis of Rivaroxaban from $MfnNH_2$ and (*R*)-ECH[a]



[a]Conditions: (a) IPA, reflux, 8 h. (b) DMF, 100 °C, 3 h. (c) DIPEA, MeCN, rt, 4 h. (d) DIPEA, DMAP, DCM, 0 °C to rt, overnight. (e) K_2CO_3, IPA, 60 °C, 15 h.

Scheme 14. Synthesis of Rivaroxaban from $MfnNH_2$ and (*R*)-ECH via the Organic Isocyanate[a]



[a]Conditions: (a) PhMe, reflux, 4 h. (b) $MgI_2{\cdot}Et_2O$, THF or DCM, rt or 50 or 80 °C, 4−8 h, N_2. (c) DMF or DMSO, 80−85 °C, 12 h. (d) EtOAc, 12 h. (e) Et_3N, DCM or EtOAc, 0 °C or rt, 1−3 h.

agent, but no yield was reported. Overall, the best yield of Rivaroxaban obtained in this sequence was 49%. The pathways in Table 6 and Scheme 10 both started with **$MfnNH_2$**, but the latter had three more steps and provided a lower overall yield compared to the former, which is therefore preferred.

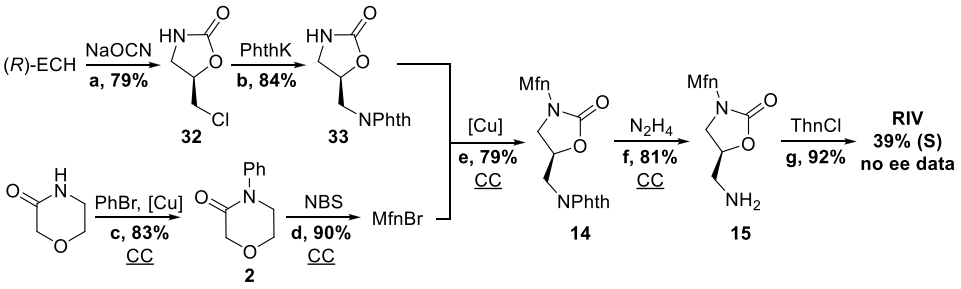
4.2. (*R*)-Epichlorohydrin, (*R*)-ECH. The first synthesis (Scheme 11) using **(*R*)-ECH** as a chiral starting material was published in 2009.[36] The target was not Rivaroxaban itself but its deuterated analogue, where all hydrogens in the morpholin-3-one ring were replaced with deuterons. Nevertheless, we think that this fact should not affect the reactions under discussion to any measurable extent and therefore included the sequence into the present review. The **(*R*)-ECH**'s ring opening with the formation of chlorohydrin **24-d_6** was conducted in a way similar to those discussed in the previous section. The yield was not reported, but after the next step (formation of the oxazolidin-2-one) the authors got only 29% of **25-d_6**. Nucleophilic substitution with potassium phthalimide gave 55% of *N*-alkyl derivative **14-d_6**, and deprotection followed by acylation afforded (through the intermediacy of amine **15-d_6**) deuterated Rivaroxaban in 11% overall yield. The reasons for the low yield are probably not in the transformations themselves but rather in the purification process because similar steps conducted by others worked better.[97,110]

Bodhuri et al. chose epoxide **28** as the key intermediate in their approach to Rivaroxaban (Scheme 12).[111] In one strategy, **$MfnNH_2$** was acylated with methyl chloroformate first, and the carbamate anion, obtained from **26a** by deprotonation with NaH, then reacted with **(*R*)-ECH** to give **28** in 40% yield (two steps). In an attempt to improve the yield, an alternative approach was tested: **(*R*)-ECH** was opened with **$MfnNH_2$**, and the resulting chlorohydrin was cyclized into the epoxide **27**. In the second stage, acylation at nitrogen with methyl chloroformate gave the same compound **28** in the same 40% yield over two steps. Then, upon action of the anion of **$ThnNH_2$** on **28** after the sequential ring opening−ring closure Rivaroxaban was obtained in 26% yield (based on

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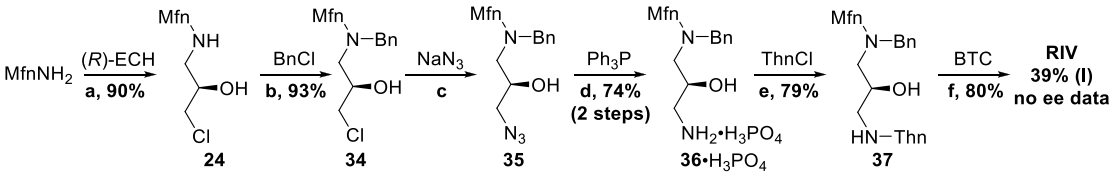
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Scheme 15. Synthesis of Rivaroxaban from $MfnNH_2$ and (*R*)-ECH via the Inorganic Cyanate[a]



[a]Conditions: (a) $MgSO_4$, H_2O, 60 °C, 1 h. (b) DMF, 80 °C, 12 h. (c) CuI, (*Z*)-1,2-diaminocyclohexane, K_2CO_3, dioxane, 110 °C, 36 h, N_2. (d) DMF, rt, 6 h. (e) CuI, (*Z*)-1,2-diaminocyclohexane, K_2CO_3, dioxane, 110 °C, 36 h, N_2. (f) MeOH, reflux, 1.5 h. (g) Et_3N, DCM, 0 °C to rt, 2.5 h.

Scheme 16. Synthesis of Rivaroxaban from $MfnNH_2$ and (*R*)-ECH via Azide Chemistry[a]



[a]Conditions: (a) IPA−H_2O, 25−35 °C, 16 h. (b) (i) DIPEA, KI, DMF, 60−70 °C, 8 h, (ii) Et_3N, 20−30 °C, 2 h. (c) wet DMF, 80−90 °C, 6 h. (d) PhMe−H_2O, from 50−60 to 25−30 °C, 6 h then H_3PO_4, MeOH, 50−60 °C, 2 h. (e) Et_3N, DCM, 25−35 °C, 2 h. (f) K_2CO_3, DCM, 25−35 °C, 8 h.

$MfnNH_2$). This yield is significantly higher than that in the previous Scheme 11 but is still relatively low.

In a subsequent work, the same authors used[97] a different approach to Rivaroxaban (Scheme 13). First, after the ring opening of **(*R*)-ECH**, they preferred to work with chlorohydrin **24** rather than cyclize it into the epoxide; subsequent nucleophilic substitution with potassium phthalimide (Scheme 13, step b) followed by acylation at the nitrogen atom with methyl chloroformate (Scheme 13, step c) proceeded in excellent yields. The secondary hydroxyl group of **29** was then acylated with **ThnCl** to give ester **30**. When treated with aqueous methylamine, the liberated amino group was acylated in an intramolecular fashion with the neighboring ester, thus giving the alcohol, which, in turn, was acylated with the carbamate. The sequence described furnished Rivaroxaban in 54% yield.

An approach (Scheme 14) to the oxazolidin-2-one ring synthesis, based on isocyanate chemistry, was published by Li et al.[54] **$MfnNH_2$** was converted into the corresponding isocyanate **MfnNCO** via the reaction with BTC followed by the magnesium iodide etherate-catalyzed reaction with **(*R*)-ECH** to give oxazolidin-2-one **25** in 55% yield (two steps). The chlorine atom was then displaced with sodium azide, and the resulting organic azide **31** was subsequently converted into corresponding amine **15** by means of catalytic hydrogenation. After the usual acylation with **ThnCl**, Rivaroxaban was obtained in 39% overall yield. The enantiomeric composition was given only as "enantiopure". In a patent,[112] the same group combined steps d and e in one pot, but that did not improve the overall yield. It should be noted that in this approach the exocyclic nitrogen atom came from an azido rather than a phthalimido group.

An inorganic cyanate was shown[113] to be useful in the oxazolidin-2-one ring assembly: the reaction between NaOCN and **(*R*)-ECH** gave heterocycle **32** in 79% yield (Scheme 15). After an S_N2 reaction with potassium phthalimide, product **33** was arylated at nitrogen with **MfnBr** (obtained via an ordinary aromatic bromination) under copper catalysis to give **14**; the same arylation conditions were employed to obtain **2** from morpholin-3-one at step c. Deprotection of **14** with hydrazine gave **15**, and its subsequent acylation with **ThnCl** afforded Rivaroxaban in 39% yield.

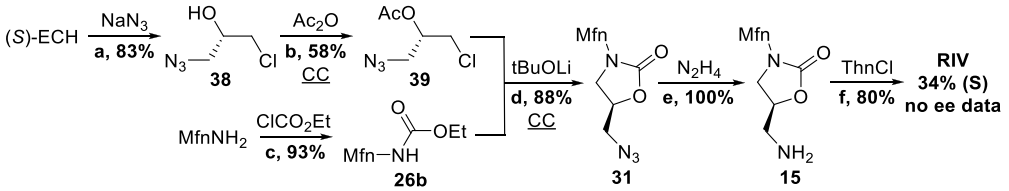
The sequence[110] depicted in Scheme 16 started with the usual reaction between **$MfnNH_2$** and **(*R*)-ECH** to afford chlorohydrin **24**. It is interesting that this transformation proceeded at a temperature slightly above rt, which is in a sharp contrast with the conditions used by others.[36,97,111] The nitrogen atom of **24** was protected with BnCl, and the chlorine atom of the resulting **34** was converted into azide **35**, which was reduced via the Staudinger reaction. Amine **36** obtained that way was isolated as its phosphate salt. Acylation of the primary amine group of **36** under standard conditions followed by one-pot deprotection of the resultant amide **37** and the oxazolidin-2-one ring assembly afforded Rivaroxaban in 39% yield.

The six synthetic sequences utilizing **(*R*)-ECH** were discussed with total yields based on **$MfnNH_2$** ranging from 11% to 54%. The synthesis in Scheme 15, which started from morpholin-3-one and produced Rivaroxaban in 39% yield, cannot be compared with the others directly, but it is not probably preferred on a large scale because of two reasons. First, the two copper-catalyzed steps with CuI loadings of 40% and 50% produced a significant and unacceptable amount of waste not to mention the associated workup difficulties. Second, morpholin-3-one is not an available starting material (see section 3.2.2). Overall, the six ways were tested on scales no more than 100 g, and in almost all of them column chromatography was used at least once. In the most high-yielding synthesis (Scheme 13), the worst step was **(*R*)-ECH**'s ring opening, while the two best steps, namely, c and d, afforded products that required chromatographic purification. If these issues are addressed, the sequence could be amenable

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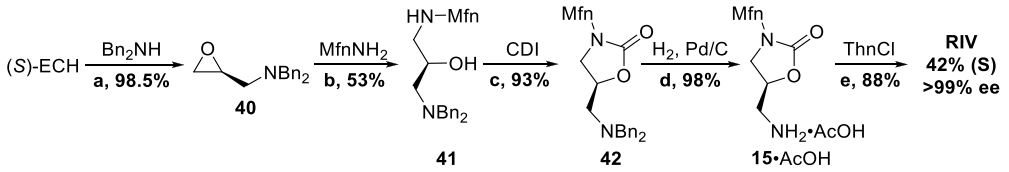
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Scheme 17. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-ECH via Azide Chemistry[a]



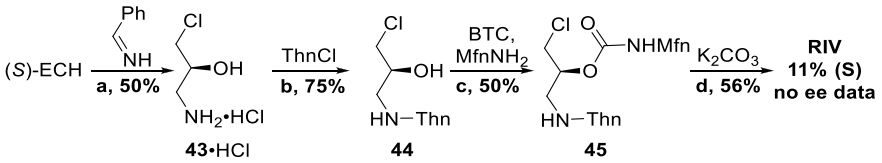
[a]Conditions: (a) H_2O−AcOH, 0 °C to rt, overnight. (b) Py, DCM, 30 °C, overnight. (c) $NaHCO_3$, Me_2CO, H_2O, 0 °C to rt, overnight. (d) DMF, N_2, 60 °C, overnight. (e) $FeCl_3$, C_{act}, EtOH, reflux, 5 h. (f) Et_3N, DCM, 0 °C to rt, overnight.

Scheme 18. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-ECH via Dibenzylamine[a]



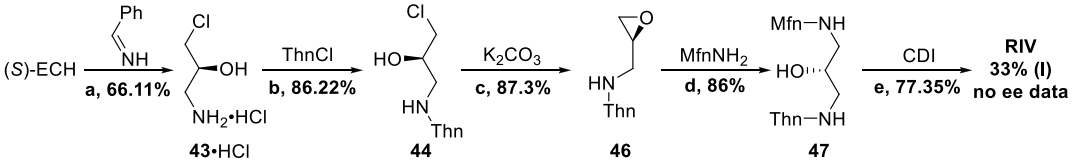
[a]Conditions: (a) (i) IPA, 0 °C to rt, 24 h, (ii) KOH, 0 °C, 30 min. (b) IPA−H_2O, 80−82 °C, 44 h. (c) PhMe, EtOH, 80−83 °C to rt, 1.5 h. (d) AcOH, rt, 24 h. (e) Na_2CO_3, H_2O, Me_2CO, PhMe, 50−55 °C, 30 min.

Scheme 19. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-ECH[a]



[a]Conditions: (a) (i) EtOH, 25−40 °C, 18 h, (ii) HCl, PhMe, 35−40 °C, 2 h. (b) (i) $NaHCO_3$, THF−H_2O, 25−30 °C, 10−15 min, (ii) PhMe, 10−15 °C, 2 h. (c) (i) Py, DCM, 10−15 °C, 1 h, (ii) Py, DMAP, rt, 2 h. (d) Me_2CO−H_2O, 25−30 °C, 12 h.

Scheme 20. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-ECH[a]



[a]Conditions: (a) (i) MeOH, 25−40 °C, 18 h, (ii) aq. HCl, PhMe, 25−30 °C, 3 h. (b) (i) $NaHCO_3$, H_2O, 25−30 °C, 30 min, (ii) **ThnCl**, PhMe, 25−30 °C, 4 h. (c) DCM-DMF, 25−30 °C, 15 h. (d) $Mg(ClO_4)_2$ (cat.), MeCN, 25−30 °C, 3 h. (e) CDI, EtOAc, 25−30 °C, 8 h.

for larger scale trials. The yield of 54% (Scheme 13) is comparable (of course, if it remains the same on a larger scale) with those obtained when (*S*)-**GPI** (Table 6) was used as a chiral starting material, considering the fact that the reagent was obtained separately. In addition, no ee data was provided in all but one work, and even in that one words only were given and not a percentage. In the case given in Scheme 16 a speculation could be made that the ee value was similar (because of the same chemical nature of the stereochemistry-determining step) to those obtained with (*S*)-**GPI** as a starting material (Table 6). In the other two cases (Scheme 14 and 15) that remains difficult to assume.

4.3. (*S*)-Epichlorohydrin (*S*)-ECH. The three-membered ring of (*S*)-**ECH** was opened with sodium azide (Scheme 17),[114] and the hydroxyl group of resulting compound **38** was acylated to give **39**. It was used further as an alkylating agent for **26b** (obtained via a reaction between $MfnNH_2$ and ethyl chloroformate) with the formation of azide **31**. After its reduction and acylation at the nitrogen atom, Rivaroxaban was obtained in 34% overall yield. The epoxide ring opening and the acylation with acetic anhydride are two relatively inefficient steps, which negatively affected the total yield.

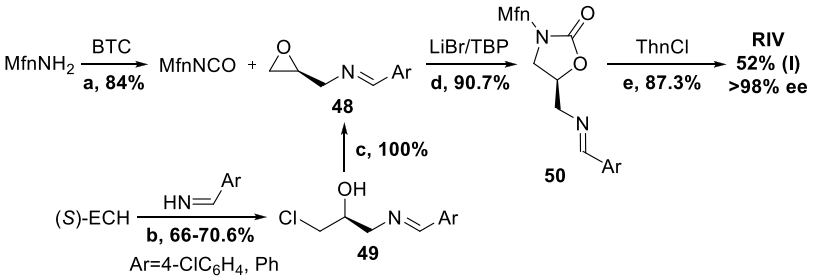
The reaction between (*S*)-**ECH** and dibenzylamine gave epoxide **40** in nearly quantitative yield; however, its opening with $MfnNH_2$ afforded only slightly more than 50% of **41** (Scheme 18).[115] The oxazolidin-2-one **42** synthesis with CDI followed by hydrogenative removal of both benzyls (to afford **15** as its acetate) and acylation of **15** gave Rivaroxaban in 39% overall yield. The enantiomeric composition was assessed on several stages, and the ee was always greater than 99%. The yield is moderate, mostly because of step b, which was both low yielding and very time and energy inefficient.

In this case (Scheme 19),[116] (*S*)-**ECH** was opened with benzaldimine, formed in situ from benzaldehyde and ammonia. The resulting imine was hydrolyzed under acidic conditions to give amine hydrochloride **43**·HCl that was acylated with **ThnCl**. The mixed carbonate (from **44** and BTC) reacted with $MfnNH_2$, and the resulting product **45** was rearranged into Rivaroxaban in base. Yields in all steps were modest, and the overall yield obtained was only 11%.

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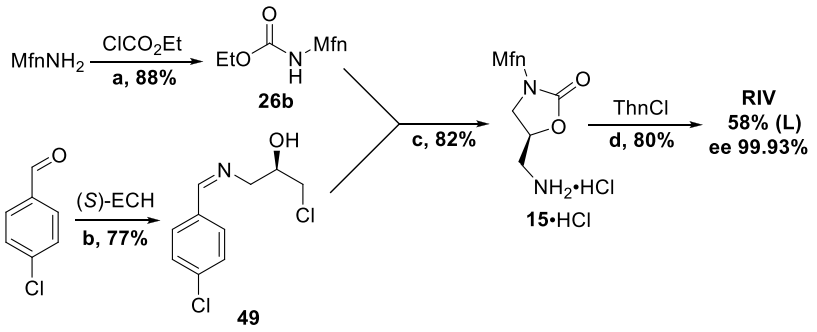
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Scheme 21. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-ECH[a]



[a]Conditions: (a) PhMe, Ar, reflux, 6.5 h. (b) EtOH, 40 °C to rt, 18 h. (c) K_2CO_3, MeOH, rt, 2 h. (d) PhMe, Ar, reflux, 1.5 h. (e) (i) HCl, DCM−H_2O, rt, 1.5 h, (ii) aq. NaOH, pH = 9−10, (iii) **ThnCl**, DCM, 0 °C to rt, 30 min.

Scheme 22. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-ECH[a]



[a]Conditions: (a) K_2CO_3, PhMe−H_2O, 0−5 °C to rt, 3 h. (b) MeOH, aq. NH_3, 25−40 °C, 14 h. (c) (i) *t*BuOLi, DCM, reflux, 12 h, (ii) EtOH, HCl, pH = 1−2, rt, 1.5 h. (d) Na_2CO_3, H_2O, PhMe, 8−12 °C to rt, 2 h.

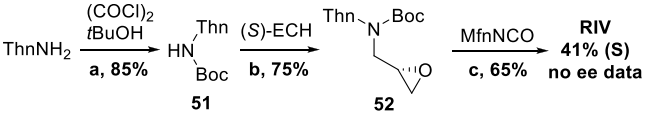
In a subsequent patent,[117] the same group was able to raise the total yield and to test the procedure on a larger scale (Scheme 20). The first two stages were the same as in the previous work,[116] but the yields became substantially higher. Next, chlorohydrin **44** was treated with a base to give epoxide **46** that was then opened with **$MfnNH_2$**. Finally, the oxazolidin-2-one ring closure was achieved by the treatment of **47** with CDI. Though the authors managed to triple the overall yield, it was still relatively low, mostly because of the initial ring opening (step a).

Another synthetic sequence (Scheme 21) where benzaldimines (formed in situ) were used as nucleophiles for (*S*)-**ECH**'s ring opening employed the Lewis acid-catalyzed reaction between isocyanate **MfnNCO** and epoxide **48** as a key step.[118] A LiBr/TBP (tributyl phosphate) system promoted the reaction with high efficiency (>90% yield of **50**). The imine obtained was hydrolyzed, and the resulting amine was acylated to afford Rivaroxaban in 52% yield.

One more related synthesis (Scheme 22), where the (*S*)-**ECH**'s three-membered ring was opened with benzaldimine, was published by Tian and co-workers.[95] In this case, alkylated imine **49** was neither hydrolyzed immediately nor cyclized into the epoxide (though it is likely to form in situ under the reaction conditions) but used as an alkylating agent for carbamate **26b**, which was in turn obtained from **$MfnNH_2$** and ethyl chloroformate. Then, the immediate imine product was hydrolyzed, and **15**·HCl was isolated by filtration, while 4-chlorobenzaldehyde was recycled from the mother liquor. After the usual acylation with **ThnCl**, Rivaroxaban was obtained in 58% yield and >99.9% ee. Apart from the decent yield, the procedure is noteworthy because of its scale of >1 kg. In a previous work, the same authors also tested[119] other conditions for each step on a small scale, and they were able to raise the overall yield up to 81%, but it seems that these conditions could not be scaled without a yield depression. In a very recent work,[120] the same pathway as that shown in Scheme 22 was followed with only minor distinctions to give Rivaroxaban in 53% yield (from (*S*)-**ECH**) with 99.94% ee on more than a 1 kg scale. It is interesting that Tian et al.[95] discarded the benzyl carbamate **26c** (an analogue of **26b**) because it gave only 28% of **15**·HCl, while Halama et al.[120] managed to obtain 84% under essentially the same reaction conditions.

The synthetic sequence[108] given in Scheme 23 differs significantly from all of the others discussed so far because it

Scheme 23. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-ECH[a]



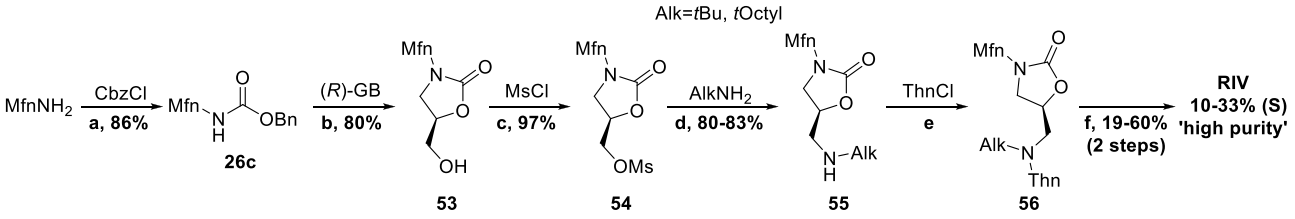
[a]Conditions: (a) (i) 1,2-DCE, 60 °C, 1 h, (ii) *t*BuOH, rt, 2 h. (b) (i) TEBAC (cat.), Na_2CO_3, IPA, 40 °C, 20 h, (ii) *t*BuOK, 5−10 °C, 2 h. (c) LiBr, TBP, PhMe, N_2, reflux, 10 h.

begins with **$ThnNH_2$**. This primary amide was additionally protected by acylation and secondary amide **51** after that was alkylated with (*S*)-**ECH** (Scheme 23) in the presence of a relatively weak base to give the corresponding chlorohydrin, which was not isolated but converted into epoxide **52** upon treatment with a stronger base. The epoxide mentioned was reacted with isocyanate **MfnNCO** under the LiBr/TBP system catalysis to afford Rivaroxaban in 41% yield.

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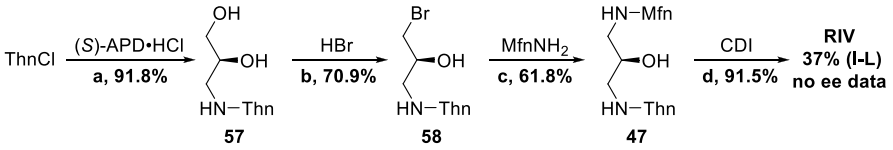
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Scheme 24. Synthesis of Rivaroxaban from $MfnNH_2$ and (*R*)-GB[a]



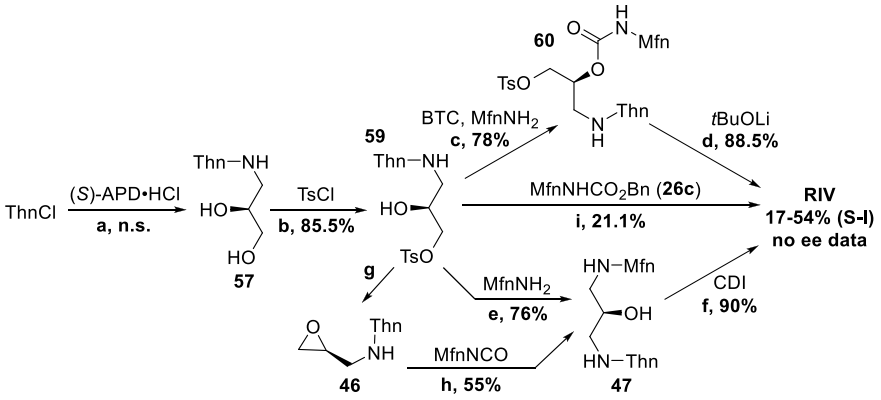
[a]Conditions: (a) $NaHCO_3$, Me_2CO-H_2O, 0 °C to rt, 4 h. (b) *t*BuOLi, THF, rt, 24 h. (c) Et_3N, DCM, 0 °C to rt, 4 h. **d** IPA, 120 °C (sealed tube), 12−120 h. (e) Et_3N, DMAP (cat.), PhMe, 110−120 °C, 2−10 h. (f) 6 M HCl, 55 °C, 2−4 h.

Scheme 25. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-APD[a]



[a]Conditions: (a) $NaHCO_3$, 2-MeTHF−H_2O, PhMe, 15−18 °C, 2 h. (b) (i) AcOH, Ac_2O, 60−65 °C, 3 h, (ii) MeOH, reflux 2.5 h, then 20−25 °C, overnight. (c) Collidine, PhMe, EtOH, 103−105 °C, 6 h. (d) PhMe-NMP, 80−115 °C, 1.5 h.

Scheme 26. Synthesis of Rivaroxaban from $MfnNH_2$ and (*S*)-APD[a]



[a]Overall yields were calculated assuming the yield of step a is 91.8% because the authors[122] made a reference to the previous procedure.[121] Conditions: (a) $NaHCO_3$, 2-MeTHF−H_2O, PhMe, 15−18 °C, 2 h. (b) (i) $(Bu)_2SnO$ (cat.), MeCN, reflux, 1 h, (ii) TsCl, Et_3N, 22 °C, 90 min. (c) Py, DMAP (cat.), DCM, N_2, 10 to 22 °C, 75 min. (d) THF, N_2, 0 °C, 1 h. (e) Collidine, PhMe, EtOH, 105 °C, 3 h. (f) PhMe-NMP, 80−110 °C, 2.5 h, N_2. (g) K_2CO_3, DCM, N_2, rt, 72 h. (h) LiBr (cat.), TBP (cat.), PhMe, N_2, reflux, 3 h. (i) (i) Ac_2O, Py, 0 °C, 1 h, (ii) $MfnNHCO_2Bn$, *t*BuOLi, MeOH, MeCN, 0 °C, 17 h.

Of all the syntheses given in this section, only one[95] was tested on a large scale and at the same time provided the highest yield (Scheme 22). Because of the relatively high yields on each step, it is difficult to suggest major adjustments; it seems that only a thorough screening of all reaction parameters could further improve the set of conditions.

4.4. (*R*)-Glycidyl Butyrate (*R*)-GB. The approach[59] given in Scheme 24 is relatively uncommon because it uses a *tert*-alkylamine to introduce the nitrogen atom, while most published works accomplished it via the Gabriel synthesis or the azide route. **(*R*)-GB** was used as an alkylating agent for carbamate **26c** to afford corresponding oxazolidin-2-one **53**. Its hydroxyl group was esterified with mesyl chloride, and mesylate **54** was used to alkylate bulky primary amines. Despite severe steric hindrance and very long reaction times, both amines **55** were obtained in good yields. Acylation then gave tertiary amides **56**, and the deprotection under acidic conditions gave the target compound in 10−33% overall yield. Enantiomeric data was provided only as the following statement: "high purity with respect to the *R*-isomer". In summary, this pathway needs a major improvement, especially in the deprotection efficiency (step f), before it could be tried on a larger scale.

4.5. (*S*)-3-Amino-1,2-propanediol, (*S*)-APD. Only a few attempts to use **(*S*)-APD** are known in the literature.[121,122] The chiral starting material already has a nitrogen atom; therefore, both groups decided to install the thiophene fragment first (step a, Schemes 25[121] and 26[122]). A leaving group was created from the primary hydroxyl of **57** (bromide **58** in Scheme 25; tosylate **59** in Scheme 26). Then, in the first case (Scheme 25), **$MfnNH_2$** was alkylated with the bromide and the final ring closure of **47** with CDI afforded Rivaroxaban in 37% overall yield.

In the other publication[122] (Scheme 26), four different pathways were tested. The first pathway (Scheme 26, abef) was the same as that above (an S_N2 reaction between **59** and **$MfnNH_2$** gave **47**, and then CDI was used for the final cyclization) and gave 54% of Rivaroxaban. The second one

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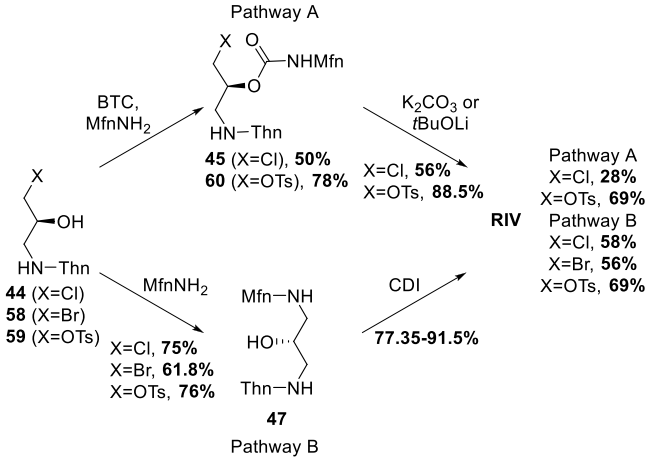
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(Scheme 26, abcd) proceeded through the acylation of the secondary hydroxyl of **59** with BTC followed by acylation of **MfnNH$_2$** with the mixed carbonate obtained. The cyclization of intermediate **60** under basic conditions afforded the target compound in the same yield of 54%. The third synthesis (Scheme 26, abghf) was a longer route to **47**: tosylate **59** was converted to epoxide **46**, which was then reacted with **MfnNCO** under LiBr/TBP catalysis. Then, the final cyclization was conducted with CDI in the usual way to give Rivaroxaban in 39% yield. The fourth route (Scheme 26, abi) was the shortest and gave the lowest yield of Rivaroxaban at the same time. Step i was comprised of a sequence of transformations conducted in one pot: first, the secondary hydroxyl of **59** was acetylated, second, the nitrogen atom in **26c** was alkylated with the resulting compound, and third, the acetyl group removal by an alkoxide was followed by immediate cyclization into Rivaroxaban.

The four-step synthesis (abef) of **RIV**, given in Scheme 26, is relatively short and provides the highest yield among the sequences starting from **(*S*)-APD**. The most problematic step in terms of yield is step e, probably because of the competition between the direct nucleophilic substitution and the epoxide formation/cleavage. If this step was improved, the entire sequence would be more attractive for a possible large-scale application because the overall yield would become comparable with that obtained in one of the best sequences discussed so far (Table 6, entry 6).

It is interesting to note some chemical similarities between the pathways given in Schemes 19, 20, 25, and 26. All of them have one common intermediate, containing a secondary hydroxyl group and a vicinal leaving group (compounds **44**, **58**, and **59**). Subsequent chemistry is also similar and can be subdivided into two pathways: A (acylation at the secondary hydroxyl followed by base mediated cyclization) and B (nucleophilic substitution followed by the oxazolidin-2-one formation with CDI). This is summarized in Scheme 27. It is

Scheme 27. Syntheses of Rivaroxaban from 44, 58, and 59 as Starting Materials



clear that regardless of the pathway, the tosyl group gave better results than halogens, and if one decides to improve these existing methodologies the best starting point is compound **59**. Since it is reasonable to obtain **59** from **(*S*)-APD**, this chiral precursor could be more useful than **(*S*)-ECH** (from which chloride **44** can be obtained) under the same conditions.

4.6. Miscellaneous. To the best of our knowledge, there is only one synthetic route where the chiral center of Rivaroxaban is created via a stereoselective synthesis rather than introduced through a chiral starting material (Scheme 28).[123] The synthesis started with a reductive amination of **MfnNH$_2$** with glyoxal dimethyl acetal followed by acylation with ethyl chloroformate and acetal hydrolysis to give key aldehyde **63**. It was further subjected to an asymmetric Henry reaction under copper catalysis, which afforded chiral compound **64** in good yield and 87% ee. The nitro group was hydrogenated and acylated as usual; after treatment with a base, the product cyclized into Rivaroxaban. Though the yield obtained was relatively modest and the ee value was not very good, this synthesis showed an example of a completely different strategy of the Rivaroxaban assembly, which might inspire other groups to put their efforts into further development of related approaches.

A few syntheses of racemic Rivaroxaban are not included in the present review because they cannot be compared directly with those giving the enantiopure or at least enantioenriched drug. The references can be found elsewhere.[28]

5. CONCLUSION

A multitude of published synthetic approaches to Rivaroxaban were discussed; their differences in the number of steps, the reagents used, and the overall yields as well as advantages and disadvantages were highlighted. Unfortunately, in the vast majority of cases, the performance was not evaluated on a large scale, so the yields published should be used carefully as a measure of the sequence efficiency. It was found that despite the diversity of approaches, two main building blocks, namely, **MfnNH$_2$** and **ThnCl**, were used very frequently. We suggested that the former can be synthesized optimally via a three-step sequence, starting from 1-halo-4-nitrobenzene. As for the latter, despite the existence of several acceptable pathways, there is still some room for improvement.

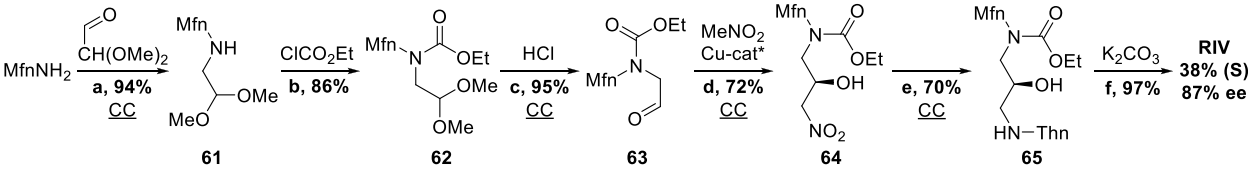
All approaches have their own advantages and disadvantages, and it is probably impossible to select the best pathway, mostly because availability, governmental regulations, and the assessment of the environmental impact of reagents and solvents vary with time and country. Nevertheless, after consideration of all of the data gathered, we infer that the most optimized sequences to date are those utilizing **(*S*)-GPI**[51] and **(*S*)-ECH**[95] as chiral starting materials. That conclusion was made by consideration of such reaction parameters such as the number of steps, yields, and chemical and optical purities, scale, and experimental protocols aggregated, and for convenience, the detailed reaction schemes are given in Scheme 29). Of course, despite the obvious advantages, some issues remain to be addressed, such as the usage of excessive reagents, the reagents themselves (the ester in the acylation step is probably worse than the acid chloride in terms of atom economy), and the replacement of solvents with greener alternatives.

We believe that new approaches to Rivaroxaban are very likely to appear in the future along with the condition adjustments for the steps of the known synthetic sequences. Because of the growing demand for more environmentally benign reagents and solvents as well as the development of new catalytic transformations, we expect those novel pathways to be safer and higher yielding than the existing ones. Moreover, the possibility of conduction of a one-pot synthesis of Rivaroxaban

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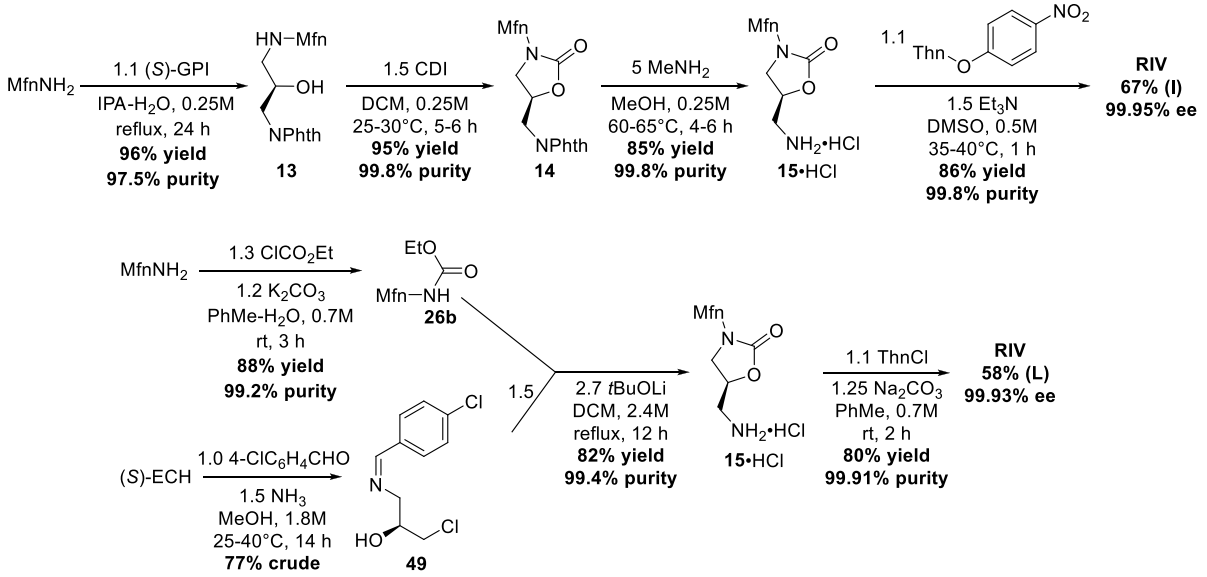
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Scheme 28. Synthesis of Rivaroxaban from $MfnNH_2$ via an Asymmetric Henry Reaction[a]



[a]Conditions: (a) (i) DCM, MS 4 Å, rt, 2 h, (ii) $NaBH(OAc)_3$, rt, 1 h. (b) Py, DCM, rt, 1.5 h. (c) MeCN−H_2O, rt, 18 h. (d) IPA, 6 °C, 168 h. (e) (i) H_2 (20 bar), Pd/C (10%), MeOH, rt, 24 h, (ii) **ThnCl**, Et_3N, DCM, rt, 3 h. (f) MeOH, rt, 2 h.

Scheme 29. Two Most Optimized and High-Yielding Syntheses



was only touched upon, and we think that work in the area is underway.

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Notes

The authors declare no competing financial interest.

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